

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Winter 2026****Course: B. Tech.****Branch : Mechanical Engineering****Semester : V****Subject Code & Name: BTMPE504A, Refrigeration and Air Conditioning****Max Marks: 60****Date: 21/02/2026****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Air refrigeration system is most suitable and commonly used in A) Ground duty applications B) Air crafts C) both D) None of the above	Understand/CO1	1
2	The practical air refrigeration systems works on A) Bellcoleman cycle B) Carnot cycle C) kalin cycle D) Rankine cycle	Understand/CO1	1
3	Compressor is replaced in vapour absorption system with A) Condenser B) Evaporator C) Absorber D) Absorber and generator	Memory/CO2	1
4	R134a refrigerant is the family of A) CFC B) HCFC C) HFC D) Natural Refrigerant	Memory/CO2	1
5	NH3 refrigerant is the family of A) CFC B) HCFC C) HFC D) Natural Refrigerant	Memory/CO2	1
6	Cascade refrigeration system uses A] only one refrigerant B] only two refrigerants C] more than one refrigerant	Memory/CO2	1

	D] All of the above		
7	In a domestic refrigerator, the condition of refrigerant at the exit of the condenser always A] dry vapour B] wet vapour C] superheated vapour D] high pressure saturated or subcooled liquid	Analysis/ CO2	1
8	The air conditioning process takes place in a domestic air conditioners in summer is A] sensible cooling B] humidification C] cooling and humidification D] Adiabatic cooling	Analysis /CO3	1
9	The most common material used for air conditioning duct manufacturing is A) AI B) GI sheet C) glass D) SS	Memory/ CO2	1
10	If the air is 100% saturated it implies that A) WBT>DBT B) WBT < DBT C) WBT = DBT D) none of the above	Understa nd/CO1	1
11	In a LiBr water absorption refrigeration system, water acts as the A) refrigerant B) absorbent C) both D) equalizer	Analysis/ CO2	1
12	Practically the Bypass factor of the CC of an air conditioner is A) >1 B) <1 C) =1 D) 0	Memory/ CO2	1
Q. 2	Solve the following.		12
A)	Differentiate between Air conditioning and Refrigeration process.	Underst and/CO 1	6
B)	Explain the simple aircraft air condition system supported with T-s representation and flow diagram.	Analysis /CO1	6
Q.3	Solve the following.		12
A)	Clearly differentiate between Dry compression and wet compression.	Applicat ion/CO2	6
B)	Show the effect of increasing the condenser pressure on CC, COP and mass flow rate supported with P-h representation.	Analysis /CO2	6
Q. 4	Solve the following		12

A)	Draw the flow diagram of the flash gas by pass two stage vapour compression system.	Underst anding/ CO2	6
B)	Explain the concept of Vapour absorption refrigeration system assisted with suitable diagram.	Analysis /CO2	6
Q.5	Solve the following.		12
A)	Enlist the desirable properties of the refrigerant.	Analysis /CO2	6
B)	Define the following psychometric properties: Degree of saturation, WBT and dew point temperature	Apply/C O3	6
Q. 6	Solve the following.		12
A)	Define the following terms: ADP, GSHF and bypass factor	Analysis /CO2	6
B)	Enlist the guidelines of the air conditioning duct design	Applicat ion/CO3	6
	*** End ***		